Exploring the Best Time Series Model for Forecasting Daily Mean Temperature in Delhi

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STAT 443: Time Series Analysis

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March 2026

Abstract

This project compares a range of forecasting methods for daily mean temperature in Delhi, using holdout RMSE as the primary criterion for model selection. Exploratory analysis indicates strong annual seasonality, nonstationarity, and substantial short-term autocorrelation, suggesting that accurate forecasting requires models that account for both seasonal structure and serial dependence. We evaluate benchmark forecasting rules, exponential smoothing methods, ARIMA models, and regression-based models with sinusoidal seasonal terms, lagged meteorological predictors, and ARMA errors. While the benchmark and exponential smoothing methods provide reasonable forecasts, and simple exponential smoothing performs better than the naive seasonal rules, seasonal ARIMA models with period 365 offer only limited improvement. In contrast, regression models with sinusoidal seasonal terms achieve noticeably stronger performance, and the inclusion of lagged wind variables further improves forecast accuracy. Among all candidate models, the best out-of-sample performance is obtained by a model with sinusoidal seasonal terms, lagged wind speed, and ARMA(4,1) errors, with a holdout RMSE of 1.6454. Diagnostic results suggest that the residual process is approximately stationary and it is a valid representation of the generative model of the time series.

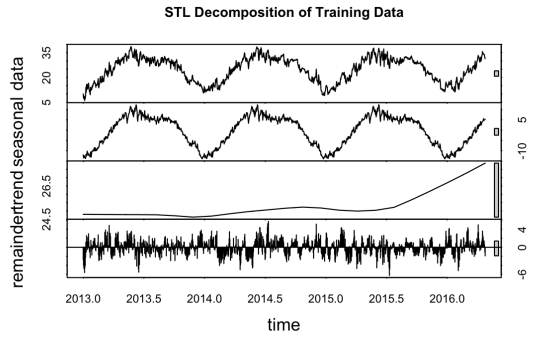
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Exploratory Data Analysis

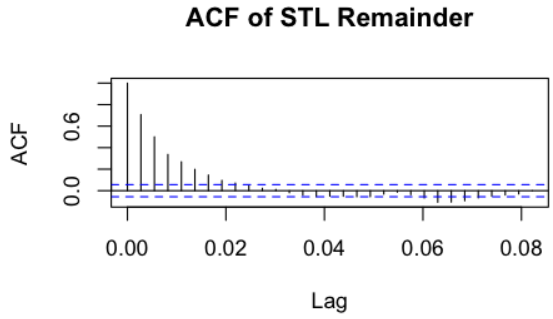
The dataset used in this project was obtained from Kaggle (2019). It contains 1,575 daily observations recorded at a weather station in Delhi, covering the period from January 2013 to April 2017. The data was observed at a daily frequency, providing approximately four years of continuous measurements. For this study, the dataset is divided into a training set (up to observation $(2016, 115)$) and a holdout set (from $(2016, 116)$ onward).

The dataset includes four climate variables: mean temperature, humidity, wind speed, and mean pressure. Mean temperature, measured in degrees Celsius, represents the daily average computed from multiple 3-hour interval readings. The sample mean temperature is 25.23°C with a standard deviation of 7.34°C. Humidity is measured in grams of water vapor per cubic meter of air (g/m^3), with a mean of 60.43 g/m^3 and a standard deviation of 16.97 g/m^3. Wind speed, measured in kilometers per hour (km/h), has a mean of 6.90 km/h and a standard deviation of 4.51 km/h. Finally, mean pressure, measured in atmospheres (atm), has a sample mean of 1011.19 atm and a standard deviation of 173.65 atm.

Since the response variable in this study is mean temperature, we first examine the raw temperature time series. All exploratory analyses presented below are conducted using the training dataset. As shown in Figure 1(a), the mean temperature exhibits clear seasonal patterns with approximately a period of 365 days. The STL decomposition in Figure 1(a) also indicates a gradual increasing trend. From the ACF plot of the remainder component from the STL decomposition, we observe that the ACF decays slowly, suggesting strong temporal dependence in the series. This behavior indicates that the series is likely nonstationary, and differencing may therefore be required when fitting models.



(a) *STL decomposition*

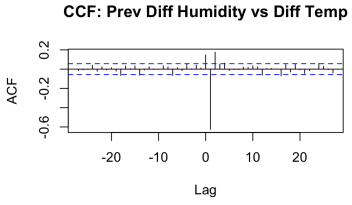


(b) *Autocorrelation plot for the residuals*

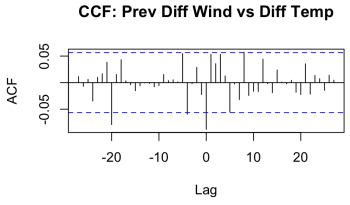
Figure 1

STL decomposition and the ACF of the residuals

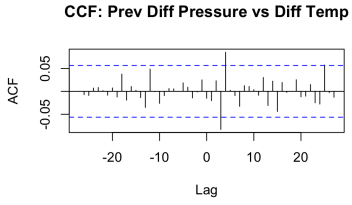
Since the dataset contains three additional climate variables—humidity, wind speed, and mean pressure—we examine the pairwise correlation plots shown in Figure A1. The plots and numerical calculations indicate a strong negative correlation between humidity and mean temperature ($\rho = -0.60$), a mild positive correlation between wind speed and temperature ($\rho = 0.31$), and a strong negative correlation between mean pressure and temperature ($\rho = -0.88$). To further assess whether these variables should be included in the forecasting model, we examine the cross-correlation plots between mean temperature and the lagged values of the other predictors. As shown in Figure 2, humidity appears to act as a leading indicator, suggesting that it may be useful to include in models such as ARMAX.



(a) *CCF Plot Between Temperature and Humidity*



(b) *CCF Plot Between Temperature and Wind Speed*



(c) *CCF Plot Between Temperature and Pressure*

Figure 2

Cross-Correlation Function (CCF) Plots Between Temperature and Other Climate Variables

Analysis Plan

To identify the best forecasting model for mean temperature, we first considered several simple benchmark rules that serve as baseline forecasting methods. These include the persistence rule, where the forecast equals the observation from the same date in the previous year, and the average rule, where the forecast is computed as the historical average of observations at the same day-of-year positions in previous years. Both benchmarks account for the strong seasonal pattern with a period of 365 days.

We then considered several exponential smoothing approaches, including simple exponential smoothing, Holt's linear trend method, and additive seasonal smoothing. In addition to these smoothing methods, we fit a range of time series models of increasing complexity. These include pure ARIMA models, regression models with sinusoidal seasonal terms, regression models with sinusoidal seasonal terms and ARIMA errors, and regression models that incorporate sinusoidal seasonal terms together with lagged covariates and ARMA errors.

For model comparison, we computed the out-of-sample RMSE using the holdout set and reported the model with the smallest RMSE as the best forecasting model for the mean temperature in Delhi. For ARIMA model fitting, we used `auto.arima` only as an initial guide because it selects models based on in-sample AIC; nearby ARIMA specifications were then manually examined and compared using the holdout RMSE.

Results and Discussion

Following the analysis plan, we consider a sequence of models of increasing complexity, beginning with benchmark forecasting rules and exponential smoothing approaches, and then moving to time series models that explicitly account for seasonal structure and serial dependence. The models we fitted, together with their out-of-sample RMSE values, are shown in Table 1, which ranks the models in ascending order of RMSE. Among the candidate models, the best forecasting performance is obtained from a regression model with sinusoidal seasonal terms, lagged wind speed as an explanatory

variable, and ARMA$(4, 1)$ errors. The parameter estimates of this final model are reported in Table 2, which provides the maximum likelihood estimates and their corresponding standard errors.

In this model, the sinusoidal terms capture the smooth annual seasonal cycle, the lagged wind variable reflects short-term meteorological influences on temperature, and the ARMA$(4, 1)$ component models the remaining temporal dependence in the residual process. This specification achieves the smallest holdout RMSE of 1.6454, making it the best forecasting model among those considered for daily mean temperature in Delhi. The final stochastic forecasting model is therefore

$$Y_t = \beta_0 + \beta_1 \sin\left(\frac{2\pi t}{365}\right) + \beta_2 \cos\left(\frac{2\pi t}{365}\right) + \beta_3 \,\mathrm{Wind}_{t-1} + R_t,$$

where the residual process follows

$$R_t = \phi_1 R_{t-1} + \phi_2 R_{t-2} + \phi_3 R_{t-3} + \phi_4 R_{t-4} + \theta_1 \varepsilon_{t-1} + \varepsilon_t,$$

with ε_t denoting white noise innovations. Again, refer to Table 2 for the parameter estimates.

Table 1
Out-of-Sample Forecast Performance of Candidate Models

Model Description	RMSE
Seasonal regression (sin/cos) + lagged wind speed + ARMA(4,1) errors	1.6454
Seasonal regression (sin/cos) + lagged wind speed and pressure + ARMA(2,1) errors	1.6506
Seasonal regression (sin/cos) + lagged humidity + ARMA(4,1) errors	1.6513
Seasonal regression (sin/cos) + ARMA(4,1) errors	1.6518
Seasonal regression (sin/cos) + lagged humidity and pressure + ARMA(3,1) errors	1.6555
Seasonal regression (sin/cos) + ARMA(4,3) errors	1.6556
Seasonal regression (sin/cos) + lagged humidity, wind, and pressure + ARMA(1,1) errors	1.6887
Seasonal regression (sin/cos) + lagged humidity and wind + ARMA(1,1) errors	1.6889
Seasonal regression (sin/cos) + lagged pressure + ARMA(1,1) errors	1.6951
Seasonal regression (sin/cos) + lagged humidity + AR(2) errors	1.6970
Simple exponential smoothing	1.7278
Holt linear exponential smoothing	1.7604
Additive seasonal exponential smoothing (Holt–Winters)	2.1817
Seasonal average benchmark (day-of-year average)	2.9562
Seasonal persistence benchmark (same day last year)	3.2743

Table 2
Maximum Likelihood Estimates for the ARMA(4,1) Regression Model

Parameter	Estimate	Std. Error
AR(1)	1.5871	0.0556
AR(2)	-0.5694	0.0644
AR(3)	-0.0658	0.0542
AR(4)	0.0382	0.0317
MA(1)	-0.8479	0.0481
Intercept	25.3239	0.6784
Cosine	-10.0741	0.8038
Sine	-0.2222	0.8018
Wind_{t-1}	-0.0280	0.0100

Model Statistic	Value
σ^2	2.441
Log-Likelihood	-2255.8
AIC	4531.61

Moving beyond prediction accuracy, we also examine whether the selected model provides a valid generative description of the data. We first assess the stationarity assumption of the residual process obtained after fitting the regression model with sinusoidal seasonal terms and the lagged wind predictor. As shown in Figure 3, the variogram for the residual series indicates that $\hat{H}$ and $\hat{G}$ closely overlap, suggesting that the residual process is approximately stationary. This supports the use of a stationary ARMA model for the residual dynamics.

This conclusion is further supported by the autocorrelation diagnostics shown in Figure 4. Both the ACF and PACF display gradually decaying tails, which is consistent with a stationary time series. In addition, the PACF exhibits a clear cutoff around lag 4 (Figure 4(b)), providing empirical support for including an AR(4) component in the ARMA$(4, 1)$ specification.

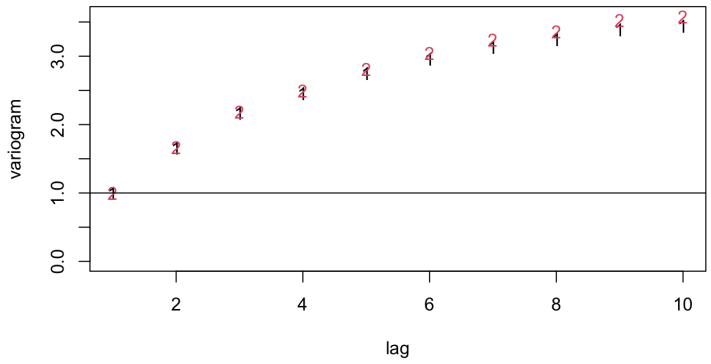
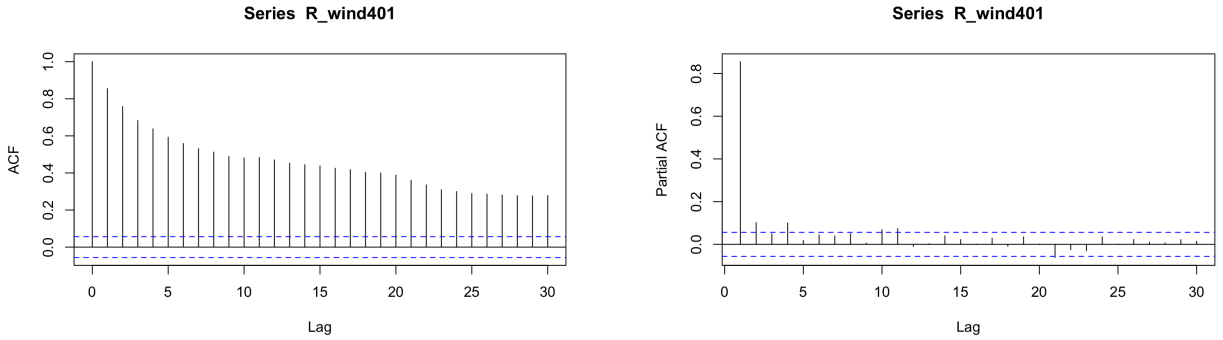


Figure 3
Variogram for the residual series of the final model

We further assess the normality assumption for the innovations in the ARMA residual process using a QQ plot of the residuals. As shown in Figure 5, the residuals lie close to the reference line, indicating that the distribution of the innovations is approximately normal. This suggests that the final model satisfies the distributional assumptions required for the ARMA error specification.

Therefore, the overall results indicate that a seasonal regression model with sinusoidal terms, lagged wind speed, and ARMA(4,1) errors provides the most accurate forecasts and an adequate stochastic representation of the daily mean temperature series in Delhi.



(a) *ACF of the residual series* (b) *PACF of the residual series*

Figure 4
Autocorrelation diagnostics for the residual series of the final model

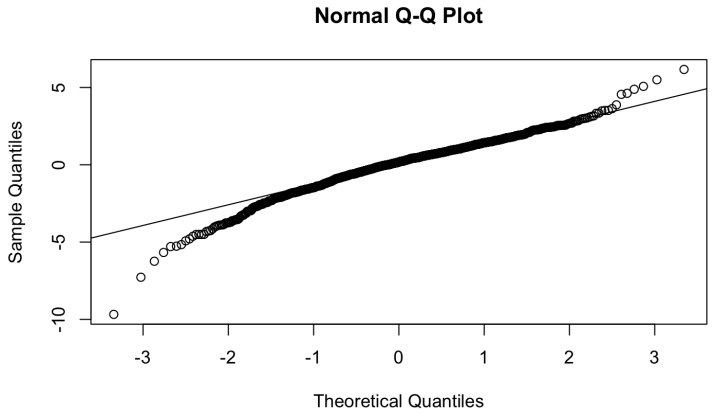


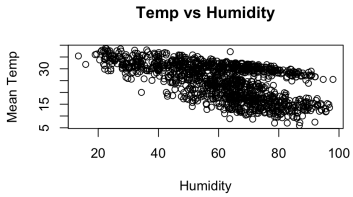
Figure 5
QQ plot of the residuals for the final model

Contribution

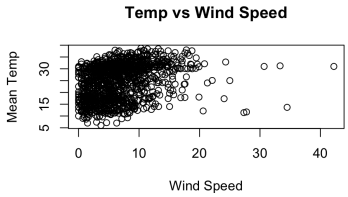
Our team was formed through the same lab. The authors are listed in alphabetical order by surname: Lu, Shi, Sowerby, and Zhu. The project was completed collaboratively, with each team member taking primary responsibility for a major section of the report.

Wei Lu took primary responsibility for coding and model implementation. Marc Shi was primarily responsible for the results and discussion section, including interpretation of model performance and diagnostic findings. Deacon Sowerby was primarily responsible for the analysis plan, including the organization and presentation of the modeling strategy. Justin Zhu was primarily responsible for writing the abstract and summarizing the overall objectives and findings of the project. All team members contributed to the overall development of the project through discussion, feedback, and revision of the final report.

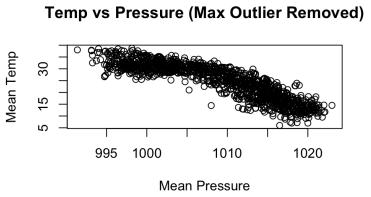
Appendix



(a) *Temperature and Humidity*



(b) *Temperature and Wind Speed*



(c) *Temperature and Pressure*

Figure A1

Pairwise Correlation Plots Between Temperature and Other Climate Variables